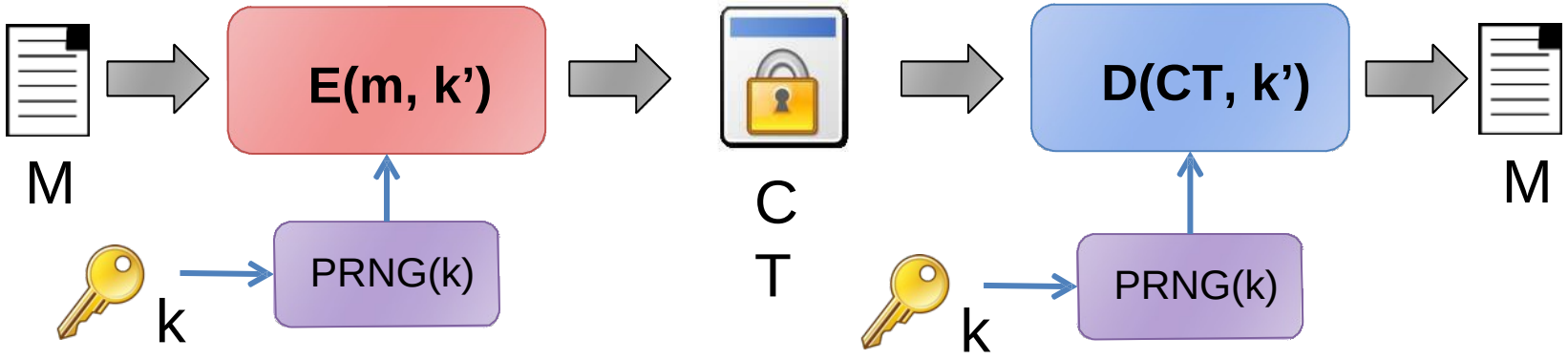
Background: stream ciphers

Practical ciphers

- **Two major categories**
  - STREAM ciphers
    - Traditional (weak): RC4
    - Modern (strong): Salsa20, ChaCha20
  - BLOCK ciphers
    - Traditional (weak): DES, 3DES
    - Modern (strong): AES
  - Plus BLOCK ciphers used in STREAM mode
    - AES-CTR

Stream ciphers



➔ **Goal: "approximate" a one-time-pad**

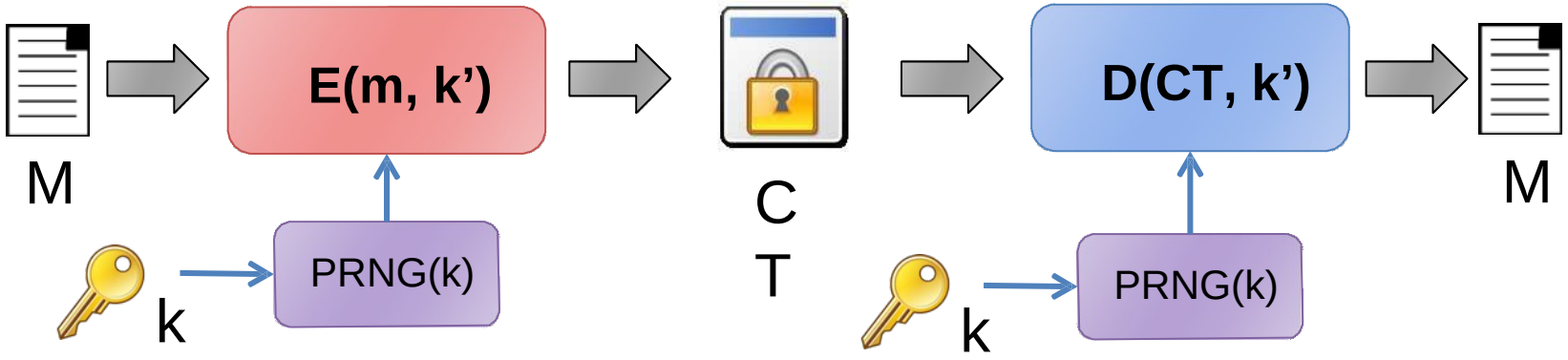
➔ **Crucial difference:**

- ⇨ One Time Pad → random key
- ⇨ Stream cipher → random keystream

➔ **DO NOT confuse key K with keystream k'!!!!**

Graphics courtesy of Marco

Stream ciphers

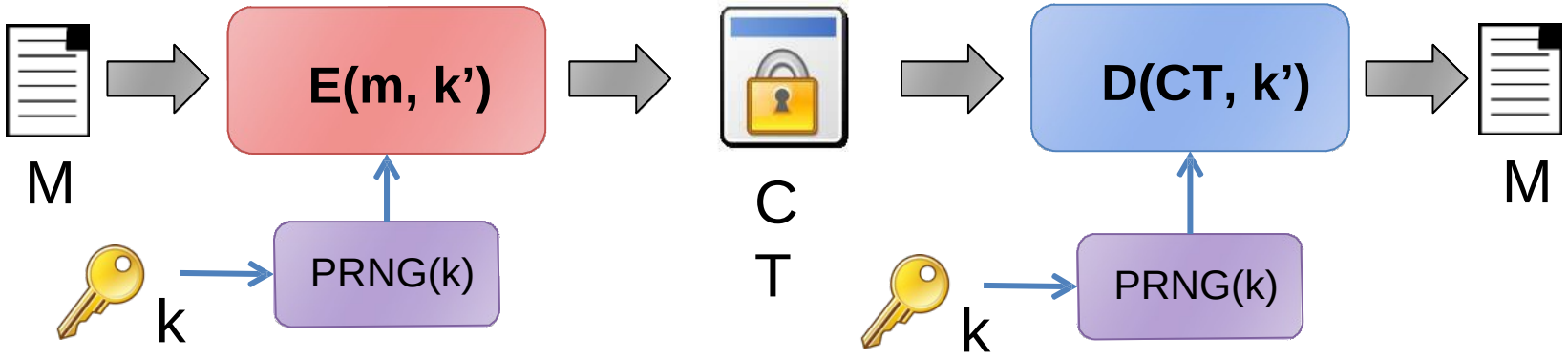


- **ENC and DEC based on XOR (exactly as OTP)**
  - CT = ENC(K, M) = M ⊕ keystream = M ⊕ PRNG(K)
  - M = DEC(K, CT) = CT ⊕ keystream = CT ⊕ PRNG(K)

- **If substring repeats, ciphertext changes (good!)**
  - C I A O C I A O C I A O ⊕
    1 3 4 f 2 5 9 5 8 d d 1 =
    α β η δ φ ϕ τ κ σ δ λ
    λ

Graphics courtesy of Marco

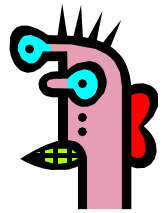
Stream ciphers

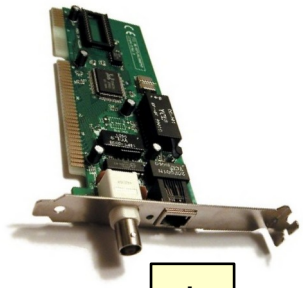


- **Still, if message is encrypted twice, ciphertext will repeat!**
  - PRNG is deterministic! same input (key K), same output (keystream K')!
- **RC4 example:**
  - Encrypt "giuseppebianchi" → CT = giuseppebianchi ⊕ RC4(key)
  - Key = 12345678790 → CT = 474d4caf78a844afa8b29add814e86
  - Key = 12345678791 → CT = d291ee1272acdb9e9e982986dfb4f8 - different key: ☺
  - Key = 12345678790 → CT = 474d4caf78a844afa8b29add814e86 - same key: ☹
- **Problem: with static key, IND-CPA breaks!**
  - Giuseppe Bianchi

Question: how to guarantee semantic security?

Initialization Vectors in stream ciphers





- ➔ **Initialization Vector IV → fresh PRNG seed at every new msg**
- ➔ **IV (usually) transmitted in clear, along with ciphertext**
- ➔ **RX combines it with long term key → reconstruct keystream**

Provable semantic security… but only if IV will NEVER REPEAT!!

Main thesis of this (entire!) class:

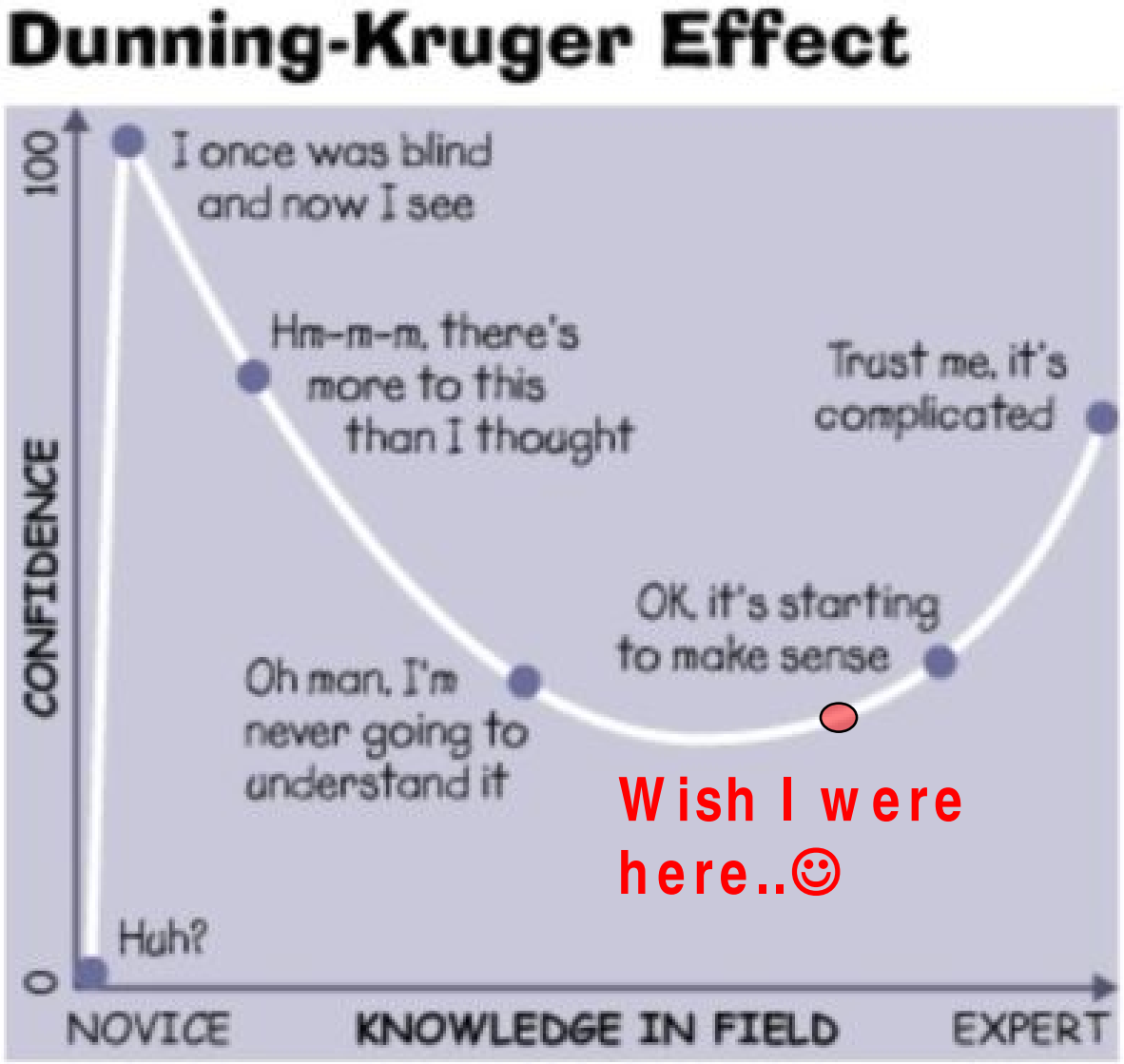
Good crypto can be badly used

Most breaches exploit poor protocol constructions and/or vulnerabilities

Warm-up example 2

802.11 WEP

Preamble: a MUST-KNOW fact of life…



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WEP lessons

- **Good cipher is far from being enough**
  - You must make good USAGE of cipher
  - And you must design GOOD protocols

- **WEP goals**
  - Authentication — Ridicolous! Facilitates attack!
  - Integrity — Inconsistent!
  - Confidentiality — Broken!

- **WEP: best "real world" example of how to make things incredibly (!) wrong**
  - And well «besides» its weak cipher (RC4)

WEP cipher: RC4

- **RC4: a specific PRNG algorithm**
  - used to generate the keystream
  - Today weak, but still considered to be strong at that time

  - **Important WEP lesson:**
    WEP confidentiality broken even if RC4 were perfect!
    - RC4 weaknesses "just" make the situation worse…

- **RC4 encryption:**
  - ENC(KEY,MSG) = MSG $\oplus$ RC4(IV, KEY)
  - Keystream = RC4(IV, KEY)
    intuitively consider this as PRNG(IV, Key)

Wep and IV

- **WEP: Per-frame IV (a must in WLAN)**
  - Stream cipher must be synchronized
    - BIG problem in lossy channels
  - WEP solution: send IV per each frame
    - Can independently decrypt each frame irrespective of previously received/missed ones

- **WEP IV: transmitted in clear**
  - Not a problem in any good stream cipher: even if IV known, this should **not** break security
    - Security should ONLY require secrecy of the key
    - Keystream = PRNG(IV,Key); if IV known key remains unknown

WEP and IV

- **If we assume RC4 strong, then WEP secure as long as IV never repeats**
- **Indeed, same keystream iff same (Key, IV)**
  - $\{ks_i\}$ = PRNG(IV,Key)
  - If IV repeats, no more semantic security
    - $(M1 \oplus ks_i) \oplus (M2 \oplus ks_i) = M1 \oplus M2$
    - If one message known, other known
- **If IV repeats, the weakness is PRACTICAL!!**
  - Easy to create a KPA or CPA condition in wifi
  - Even without KPA, Limited message entropy → helps to infer
  - "context" information → helps to infer

And since IV is soooooooo important….

➔WEP incredible blunders:

1. IV size = 24 bits

2. IV generation:
   left to the implementation
   [never, NEVER, do this in security protocols!!]

Repeating IV in WEP

- **Cyclic generation {1,2,3,…}?**
  - 24 bit → 2^{24} cycle = 16.777.216 frames
  - Assume:1500 bytes frames, ~7 mbps (net) wifi throughput
  - IVs re-cycle after less than 8h!

- **Random IV generation?**
  - Birthday paradox!
  - 50% collision probability after approx 2^{12} = 4000 frames!

- **Restart after reboot?**
  - In many cases, IV re-start from 0! Same IV sequence!!

Practical attacks to small IV space

- **Passive attack: create dictionary**
  IV → RC4(IV,Key)
  - How to? Use known messages to recovery keystream
    (MSG ⊕ RC4(IV,key)) ⊕ MSG → RC4(IV,key)
  - Known messages?
    - Send an email with large known attachment ☺
    - Use…. AUTHENTICATION PROCEDURE!!
      - **See next 4-5 slides!!**
- **Wait for IV to repeat: game over**
  (M' ⊕ RC4(IV,key)) ⊕ RC4(IV,key) = M'
- **Larger WEP key size (e.g. 128)?**
  - Irrelevant: IV still 24 bits!! Same attack time!!

Well, things get even worse
weak chosen cipher

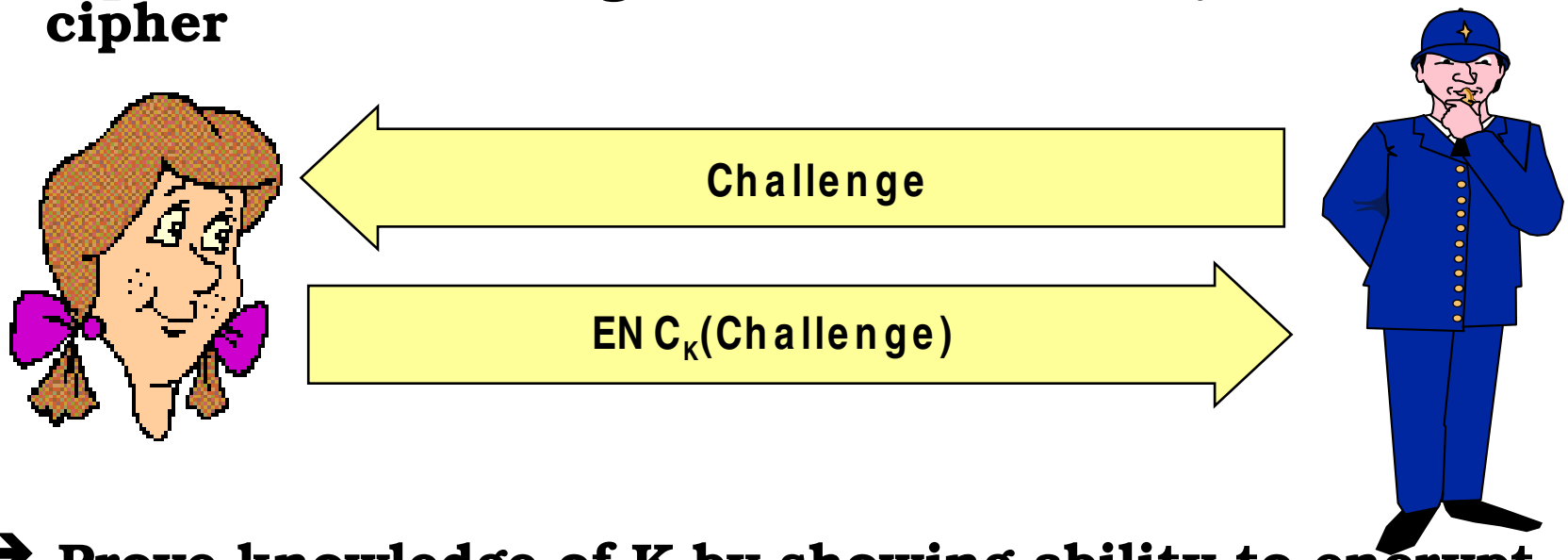
- **Cryptoanalytic attack to RC4**
- **Fluhrer, Mantin, Shamir: show weakness**
  - Some “weak” IVs leak key information into keystream
    - With 5% probability, a byte in the keystream is equal to a byte in the key
    - Look at many packets and see where bias is
- **Stubblefield, Ioannidis, Rubin use weakness**
  - Focus on first keystream byte
  - BUT this is even known for ALL packets
    - **Remember: 802.11 uses SNAP LLC encapsulation: first byte = 0xAA**
    - **Known plaintext!**
  - Attack linear (!) with key size

WEP authentication

- **Grant network access only to users who KNOW a pre-shared secret**
  - A DIFFERENT security goal (service) than confidentiality!
- **How to "prove" I know a secret?**
  - By revealing it while accessing the network?
    - Nah…
      - anyone else would overhear it
      - And would use it later on (replay attack)
  - Perform some operation which REQUIRES knowledge of the secred
    - **But whose result does NOT reveal it!**

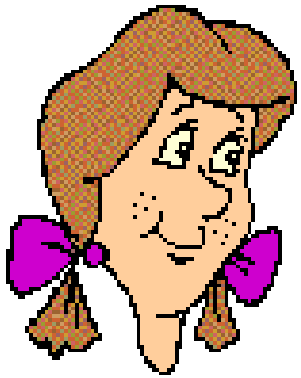
WEP idea:

- **Secret: use the SAME of encryption**
  - Minimize «secrets» to handle
    - In principle might even make sense…
      - But you MUST know how to properly handle this more in a month from now!
- **Operation: Challenge-Handshake with symmetric cipher**



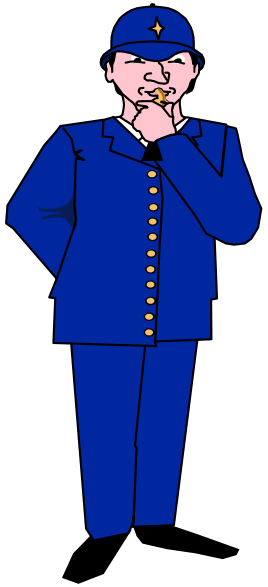
- **Prove knowledge of K by showing ability to encrypt the challenge**

It is a good approach?



Challenge

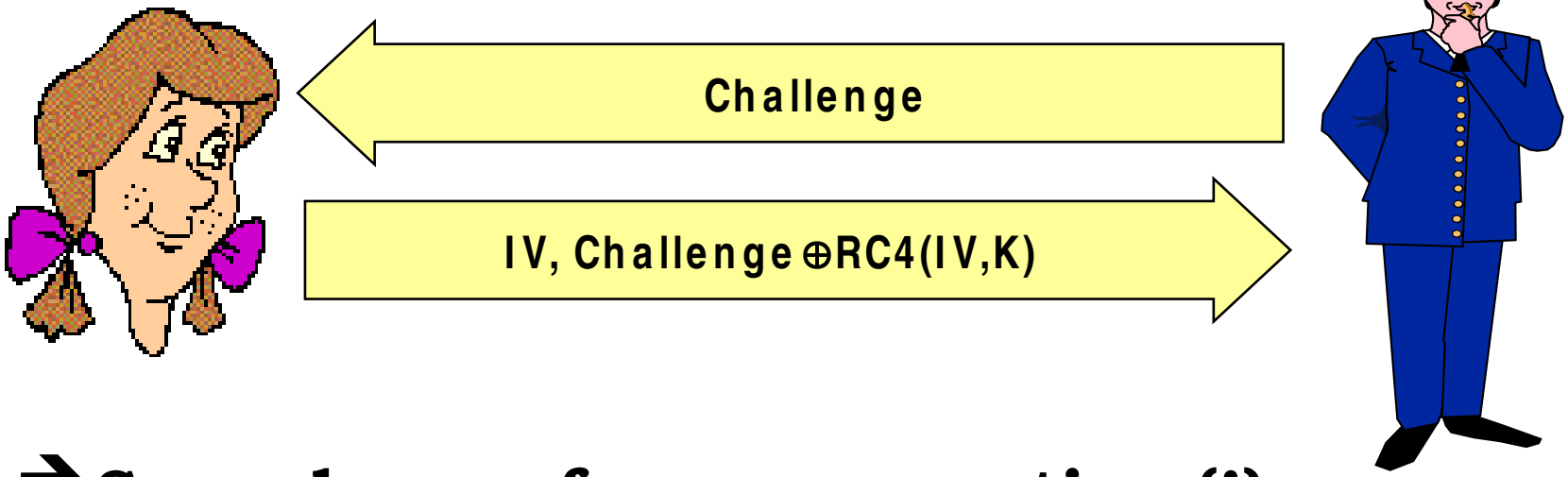
EN C_K(Challenge)



- ➔ **Let's check the Menezes book:**
  - ⇨ Yes, chapter 10.3.2 mentions this
  - ⇨ As long as challenge random and never repeats, I'm OK then
- ➔ **SURE???**

Why WEP "authentication" helps?

➔ **WEP details**



➔ **Same key as frame encryption (!)**

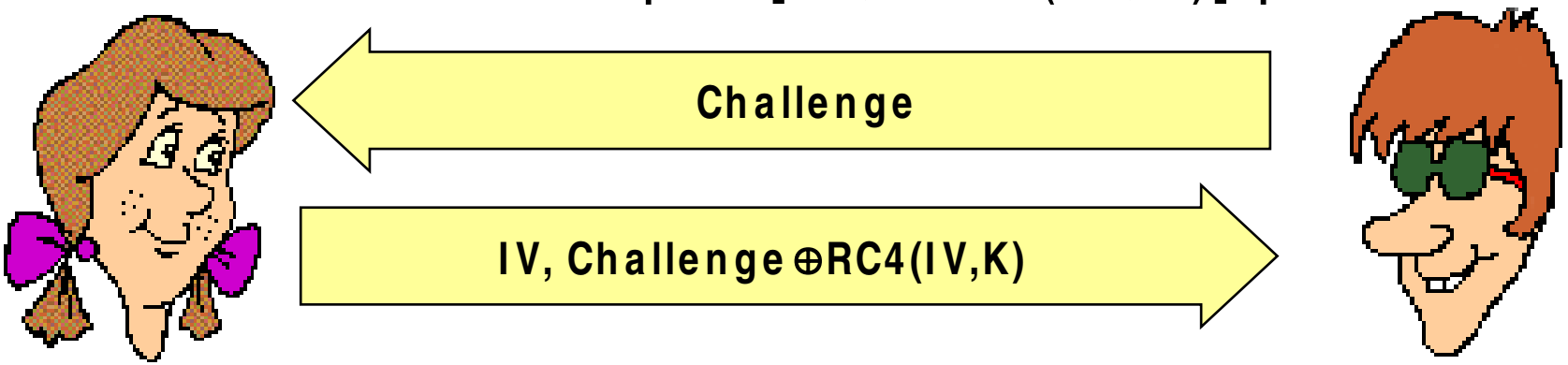
➔ **Challenge = plaintext = known**

➔ **Isn't this a Known Plaintext Attac**

Why WEP "authentication" helps?

➔**Rogue AP: send multiple challenges**

⇨ Gather multiple [IV, RC4(IV,K)] pairs!!



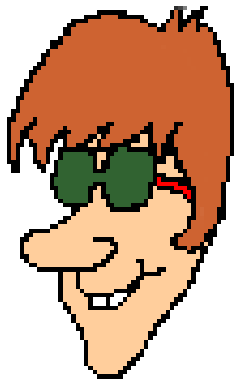
⇨ When enough, go and enjoy decrypting anyone else in the BSS!

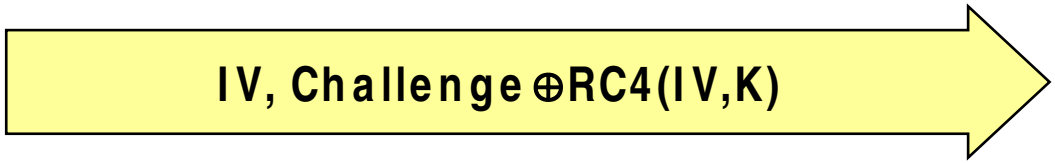
But at least is WEP "authentication" robust??

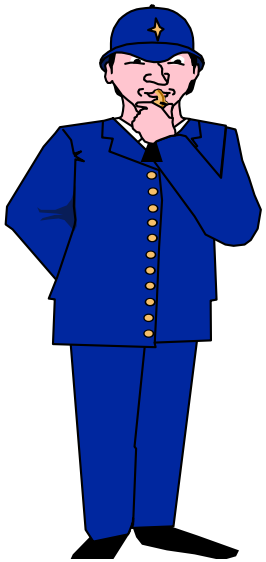
➔ **No, its broken as well!!**

No need to know key to enter network!

⇨ JUST one [IV, RC4(IV,K)] pair → game over!





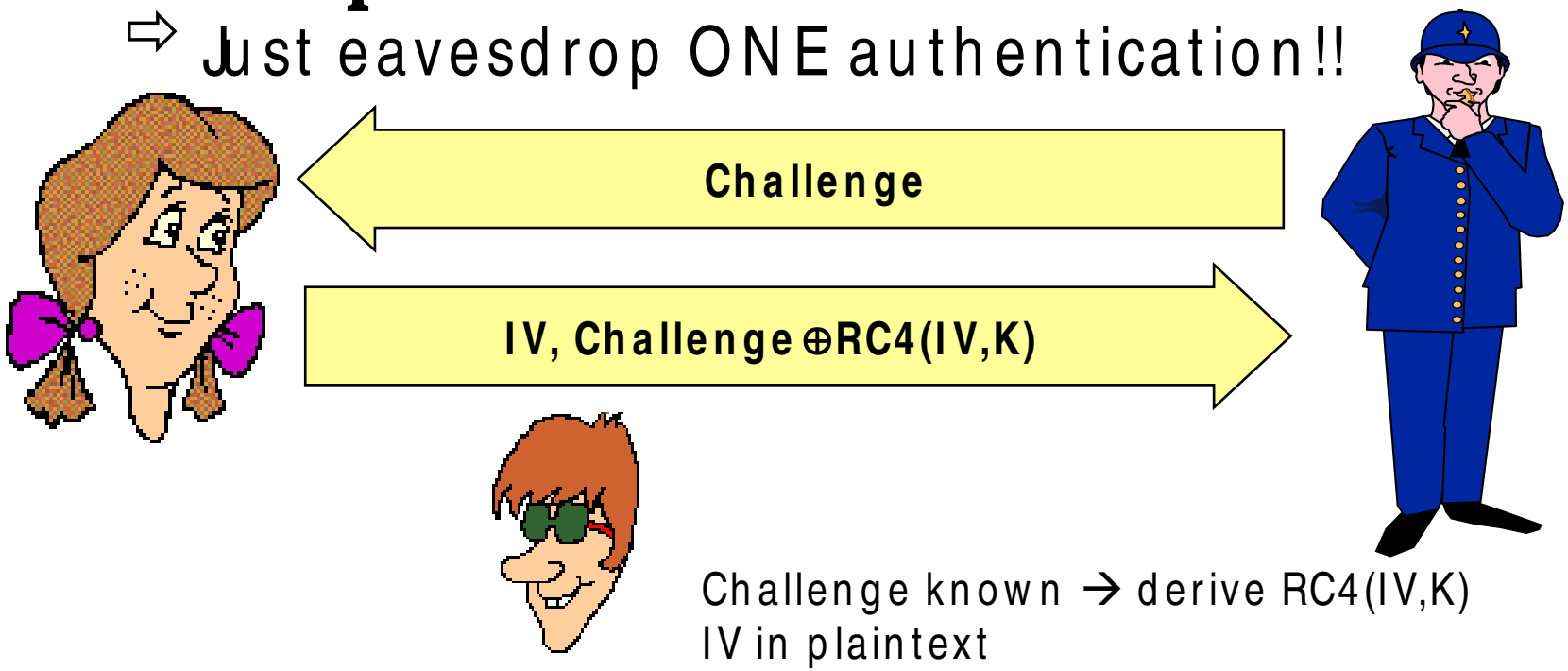


⇨ Why?

⇨ Because... IV chosen by... responde

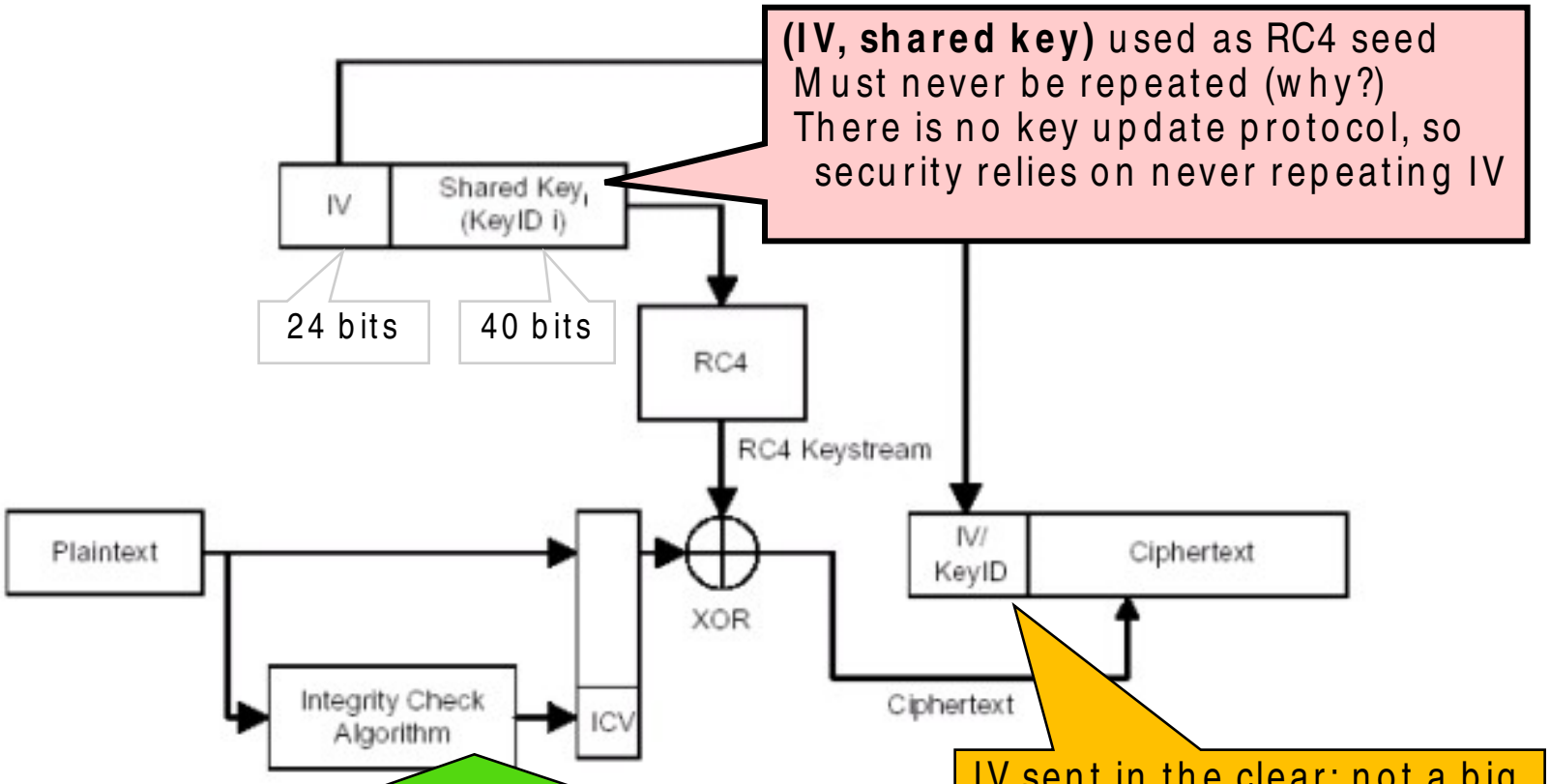
Why WEP "authentication" helps?

- **And if no IV available and no active attack possible?**
  - ⇨ Just eavesdrop ONE authentication!!



Challenge known → derive RC4(IV,K)
IV in plaintext
One pair ready to be used for YOUR authentication!

WEP – summary (so far)



no integrity!

Source: V. Shmatikov

What about integrity?

- **"Terrific" idea: use native CRC-32 as integrity check value**
  - External encryption wrapping should prevent (!) attacker to perform valid alterations
- **Problem: CRC-32 is linear!**
  - $c(A) \oplus c(B) = c(A \oplus B)$
  - **Deadly**: use only cryptographically strong algorithms
- **Consequences:**
  - (Meaningful!) Message modification
  - Message injection

Message modification

IV; C = [M | C(M)] ⊕ RC4(IV,K)

Chooses δ as long as M

Computes $C' = C \oplus \{\delta, c(\delta)\}$

Computes:

$$= \left[RC4(IV,K) \oplus \{M, c(M)\}\right] \oplus \{\delta, c(\delta)\} =$$

$$= RC4(IV,K) \oplus \{M \oplus \delta, c(M) \oplus c(\delta)\} =$$

$$= RC4(IV,K) \oplus \{M', c(M \oplus \delta)\} =$$

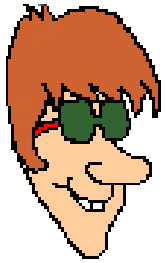
$$= RC4(IV,K) \oplus \{M', c(M')\}$$

C' is an encrypted message with valid CRC32!! Fooled!

NO NEED to know M

δ can be chosen so as to flip specific bits

Message injection



Obtain one pair IV, RC4(IV,K)

Goal: inject message M so that it is authenticated and encrypted

Steps:
- **Compute c(M)**
- **inject IV, RC4(IV,K) ⊕[M | c(M))**

Take-home messages:

1) integrity check must be NON LINEAR

2) Integrity check must explicitly include a key

external encryption may not provide ANY guarantee

802.11: Aftermath

- **Optimized WEP attacks on many open source**
  - Although just very few WEP protected networks remaining
- **WiFi evolution (802.11i):**
  - WPA: retain hardware compatibility (RC4), but patch with:
    - Longer IV (48 bit)
    - Protection of IV (plaintext differs from RC4 input)
    - Ephemeral key derivation (changing in time)
    - …
- **WPA2: as above but using AES (new hardware)**
- **2017: discovered KRACK attack to WPA2**
    - Note: attack works against a **mathematically proven secure** (!) 4-way handshake… OUCH!
- **2018+ → WPA3 standardized**